

SCX

State-Conditioned eXpertise (SCX) —



A. (Conditional Risk)

A.1 Kolmogorov

SCX $R_m(s) = \mathbb{E}_{x \sim P(\cdot|s)}[\ell(f_m(x), f^*(x))]$ Kolmogorov

A.1 (Kolmogorov, 1933). $(\Omega, \mathcal{F}, P)$ Y $\mathcal{G} \subseteq \mathcal{F}$ σ - $\mathbb{E}[Y | \mathcal{G}]$ a.s. $\mathcal{G}$ -

$$\int_A \mathbb{E}[Y | \mathcal{G}] dP = \int_A Y dP, \quad \forall A \in \mathcal{G}.$$

SCX $X \sim s \mathcal{X} \{X \in s\} \sigma$ - $\sigma(\{X \in s\})$ $R_m(s) \mathbb{E}[\ell(f_m(X), f^*(X)) | X \in s]$

$$\begin{aligned} R_m(s) &= \frac{1}{P(X \in s)} \int_{X \in s} \ell(f_m(x), f^*(x)) dP_X(x) \\ &= \frac{\mathbb{E}[\ell(f_m(X), f^*(X)) \cdot \mathbf{1}\{X \in s\}]}{P(X \in s)}. \end{aligned}$$

$P(X \in s) = 0$ $R_m(s) \lim_{\epsilon \rightarrow 0^+} R_m(s_\epsilon)$ $s_\epsilon \in s$

A.2

$r(x) = \mathbb{E}[Y | X = x]$ $\ell(\hat{y}, y) = (\hat{y} - y)^2$

$$\begin{aligned}
R_m(s) &= \mathbb{E}[(f_m(X) - f^*(X))^2 \mid X \in s] \\
&= \mathbb{E}[(f_m(X) - r(X) + r(X) - f^*(X))^2 \mid X \in s] \\
&= \underbrace{\mathbb{E}[(f_m(X) - r(X))^2 \mid X \in s]} + \underbrace{\mathbb{E}[(r(X) - f^*(X))^2 \mid X \in s]} \\
&\quad + 2 \underbrace{\mathbb{E}[(f_m(X) - r(X))(r(X) - f^*(X)) \mid X \in s]}.
\end{aligned}$$

$$f^* f^*(x) = \mathbb{E}[Y \mid X = x] f^*$$

$$R_m(s) = \mathbb{E}[(f_m(X) - r(X))^2 \mid X \in s] + \mathbb{E}[\text{Var}(Y \mid X) \mid X \in s].$$

- s s

A.3 Bayes

$$R^* = \inf_f \mathbb{E}[\ell(f(X), Y)] \mid s$$

$$R^*(s) = \inf_f \mathbb{E}[\ell(f(X), Y) \mid X \in s].$$

$$\text{SCX } R_m(s) \mid f_m \mid s \mid f^* \mid R^*(s) \mid f^* \neq \arg \inf_f \mathbb{E}[\ell(f(X), Y)]$$

A.1 (). f^*

$$R_m(s) = \underbrace{\mathbb{E}[\ell(f_m(X), r(X)) \mid X \in s]} + \underbrace{\mathbb{E}[\ell(r(X), f^*(X)) \mid X \in s]}.$$

A.4 PAC-Bayes

PAC-Bayes (McAllester, 1999)

$$\mathbb{E}_Q[\mathcal{R}(h)] \leq \mathbb{E}_Q[\hat{\mathcal{R}}(h)] + \sqrt{\frac{\text{KL}(Q \| P) + \log(n/\delta)}{2n - 1}}.$$

PAC-Bayes

$$\mathbb{E}_{Q_s}[\mathcal{R}(h \mid s)] \leq \mathbb{E}_{Q_s}[\hat{\mathcal{R}}(h \mid s)] + \sqrt{\frac{\text{KL}(Q_s \| P_s) + \log(n_s/\delta)}{2n_s - 1}},$$

$n_s = |\{i : x_i \in s\}|$ $s \in \mathcal{S}$ Q_s SCX $m^*(x) = \arg \min_m \sum_x \gamma_s(x) R_m(s)$ — PAC-Bayes

A.5 Domain Adaptation / Covariate Shift

$P_{\text{train}} P_{\text{test}} R_m(s)$

$$R_m^{\text{test}}(s) = \mathbb{E}_{x \sim P_{\text{test}}(\cdot | s)}[\ell(f_m(x), f^*(x))] = \frac{\mathbb{E}_{x \sim P_{\text{train}}}[\ell(f_m(x), f^*(x)) \cdot \mathbf{1}\{x \in s\} \cdot w(x)]}{\mathbb{E}_{x \sim P_{\text{train}}}[\mathbf{1}\{x \in s\} \cdot w(x)]}.$$

$w(x) = dP_{\text{test}}/dP_{\text{train}}(x)$ SCX covariate shift — $w(x)$

B. (State Partition)

B.1

B.1. $\mathcal{X} = \Pi = \{s_1, s_2, \dots, s_K\}$ 1. $\bigcup_{k=1}^K s_k = \mathcal{X}$ 2. $s_k \cap s_{k'} = \emptyset$ $k \neq k'$ 3. $s_k \in \mathcal{A}$ Borel σ -

Π σ - $\sigma(\Pi) = \bigcup_{k \in I} s_k$ $I \subseteq \{1, \dots, K\}$ $\mathbb{E}[Y \mid X \in s_k]$ $\mathbb{E}[Y \mid \sigma(\Pi)]$

B.2

B.2 (). $T : \mathcal{X} \rightarrow \mathcal{T}$ $\theta \perp X \mid T(X)$ $Y \perp X \mid T(X)$

SCX $\{s_k\}$ $Y \perp X \mid s(X)$

$$P(Y \mid X) = P(Y \mid s(X)), \quad \text{a.s.}$$

(Tishby et al., 1999).

$$\min_{\Pi: |\Pi|=K} I(X; \tilde{X}) - \beta I(\tilde{X}; Y),$$

$$\tilde{X} \stackrel{\Pi}{\rightarrow} X \stackrel{\beta}{\rightarrow} \infty \quad K \quad I(\tilde{X}; Y) \longrightarrow Y$$

B.3 Vapnik-Chervonenkis (partition classifier)

$$\mathcal{H}_{\Pi} = \{h : h(x) = \sum_{k=1}^K \alpha_k \mathbf{1}\{x \in s_k\}, \alpha_k \in \mathbb{R}\}.$$

$$\text{VC } O(K) \text{ - } K \text{ - SCX - VC bound } \tilde{O}(\sqrt{K/n})$$

B.4 RKHS

$$\mathcal{H}_K \quad f \quad f(x) = \langle f, K(x, \cdot) \rangle_{\mathcal{H}_K}$$

$$\textbf{k-means} \quad K \text{ RKHS } \phi(x) = K(x, \cdot)$$

$$\min_{\{s_k\}} \sum_{k=1}^K \sum_{x_i \in s_k} \|\phi(x_i) - \mu_k\|_{\mathcal{H}_K}^2,$$

$$\mu_k = \frac{1}{|s_k|} \sum_{x_i \in s_k} \phi(x_i)$$

” ”

B.5

Laplacian K -means

$$1. \quad W_{ij} = \exp(-\|x_i - x_j\|^2 / \sigma^2) \quad 2. \quad \text{Laplacian } L = I - D^{-1/2} W D^{-1/2} \quad 3. \quad L \quad K \\ v_1, \dots, v_K \quad 4. \quad [v_1, \dots, v_K] \quad K\text{-means}$$

(Cheeger cut)SCX

C. (Data Value)

C.1 Shapley

C.1 (Shapley, 1953). $N = \{1, \dots, n\}$ $v : 2^N \rightarrow \mathbb{R}$ S i Shapley

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} [v(S \cup \{i\}) - v(S)].$$

Shapley

1. **(Efficiency):** $\sum_{i=1}^n \phi_i(v) = v(N) - v(\emptyset)$
2. **(Symmetry):** $v(S \cup \{i\}) = v(S \cup \{j\})$ $S \subseteq N \setminus \{i, j\}$ $\phi_i(v) = \phi_j(v)$
3. **(Null Player):** $v(S \cup \{i\}) = v(S)$ $S \subseteq N \setminus \{i\}$ $\phi_i(v) = 0$
4. **(Additivity):** $\phi_i(v + w) = \phi_i(v) + \phi_i(w)$

C.2 Banzhaf

C.2 (/ Semivalue). Shapley

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} p_{|S|} [v(S \cup \{i\}) - v(S)],$$

$$\sum_{k=0}^{n-1} \binom{n-1}{k} p_k = 1 \quad p_k$$

Banzhaf (Banzhaf, 1965) $p_k = 1/2^{n-1}$

$$\phi_i^B(v) = \frac{1}{2^{n-1}} \sum_{S \subseteq N \setminus \{i\}} [v(S \cup \{i\}) - v(S)].$$

SCX NoiseScore(x_i) = $r_i / (\rho(s_i) + \varepsilon) \cdot [1 - C(s_i)]$ **Banzhaf** —

C.3 Influence Function

C.3 (Hampel, 1974). $T(F)$ F x influence function

$$\text{IF}(x; T, F) = \lim_{\varepsilon \rightarrow 0^+} \frac{T((1 - \varepsilon)F + \varepsilon \delta_x) - T(F)}{\varepsilon}.$$

$\hat{F}_n$ leave-one-out

$$\hat{\text{IF}}(x_i) \approx (n - 1)[T(\hat{F}_n) - T(\hat{F}_{-i})].$$

$$T(F) = \mathbb{E}_{(x,y) \sim F}[\ell(f(x), y)] \quad x_i \text{ Shapley } \varepsilon = 1/n \text{ Koh \& Liang, 2017}$$

$$\phi_i(v) \approx \frac{1}{n} \cdot \text{IF}(x_i; T, \hat{F}_n) + O(1/n^2).$$

$$\text{SCX } r_i/(\rho(s_i) + \varepsilon)$$

C.4 Expected Value of Information (EVOI)

C.4. x

$$\text{EVOI}(x) = \mathbb{E}_{Y|X=x}[U(\mid)] - U(\mid),$$

$$U$$

$$\text{SCX } s^* = \arg \max_s V(s) = \arg \max_s \bar{r}(s) \cdot \rho(s) \cdot L(s) \cdot [1 - D(s)] \cdot \max_m \text{SCX}_m(s)$$

$$\textbf{EVOI} \quad - \bar{r}(s) \quad s \quad - \rho(s) \quad - L(s) \quad - 1 - D(s) \quad - \max_m \text{SCX}_m(s)$$

C.5 Marginal Likelihood / Bayesian Model Selection

$$M$$

$$P(\mathcal{D} \mid M) = \int P(\mathcal{D} \mid \theta, M) P(\theta \mid M) \, d\theta.$$

$$\text{SCX } \Pi$$

$$\Pi^* = \arg \max_{\Pi} \sum_{k=1}^K \log P(\{y_i : x_i \in s_k\} \mid \{x_i : x_i \in s_k\},),$$

$$''' \, Y \, X$$

$$\sum_k |s_k| \cdot \hat{I}(X; Y \mid s_k),$$

D. (Expert Reliability)

D.1 Dawid-Skene

D.1 (Dawid & Skene, 1979). $N \ M$

$$z_j \in \{1, \dots, C\} \quad y_{ij} \quad i \quad \pi_{cc'}^{(i)} = P(y_{ij} = c' \mid z_j = c)$$

$$P(Y \mid \pi, z) = \prod_{j=1}^N P(z_j) \prod_{i=1}^M \pi_{z_j, y_{ij}}^{(i)}.$$

EM $\pi^{(i)} \ z_j$

SCX $SCX_m(s) = P(\ell(f_m(x), y) < \tau \mid x \in s)$ Dawid-Skene f_m ”” $SCX_m(s) \ \tau$
 SCX ——— $s \ SCX_m(s)$

D.2

$$P(y_{ij} = c' \mid x_j, z_j = c, \theta_i) = f(x_j, c, c'; \theta_i),$$

$$\theta_i \ i \ - \ : \ P(y_{ij} = z_j \mid x_j, \theta_i) = \theta_i \ - \ : \ P(y_{ij} = c' \mid z_j = c, \theta_i) = \theta_{i, cc'} \ - \ : \\ \theta_i(x) = \sigma(w_i^\top x)$$

SCX $SCX_m(s)$ ——— s

D.3 Item Response Theory (IRT)

IRT Lord, 1952; Rasch, 1960”””” (2PL)

$$P(Y_{ij} = 1 \mid \theta_j, a_i, b_i) = \sigma(a_i(\theta_j - b_i)),$$

$$\theta_j \ j \ a_i \ b_i$$

SCX $SCX_m(s)$ IRT - f_m ”” - x ”” - s - $SCX_m(s)$

IRT (MIRT)

D.4 Calibration / ECE

(Expected Calibration Error)

$$\text{ECE} = \mathbb{E}_{\hat{P}}[|P(Y = 1 \mid \hat{P} = p) - p|].$$

SCX $SCX_m(s)$ f_m s

$$\text{ECE}_m(s) = \left| \frac{1}{|s|} \sum_{x_i \in s} \mathbf{1}\{\ell(f_m(x_i), y_i) < \tau\} - \mathbb{E}_{x \in s}[SCX_m(s)] \right|.$$

$\tau \rightarrow 0$ $SCX_m(s)$ s

D.5 Conformal Prediction

(Vovk et al., 2005) $1 - \alpha$

$$\Gamma^\alpha(x) = \{y : p_y(x) > \alpha\}, \quad p_y(x) = \frac{|\{i : s(x_i, y) \leq s(x, y)\}| + 1}{n + 1}.$$

SCX τ α $SCX_m(s)$ τ f_m s

D.1. s τ

$$\mathbb{E}[SCX_m(s)] = 1 - \alpha,$$

α τ

E. (Noise & Learnability)

E.1 Tsybakov

E.1 (Tsybakov, 2004). $\eta(x) = P(Y = 1 \mid X = x)$ P Tsybakov $\kappa \geq 0$
 $C > 0$

$$P_X(|\eta(X) - 1/2| \leq t) \leq Ct^\kappa, \quad \forall t > 0.$$

$$\kappa - \kappa = 0 - \kappa \rightarrow \infty$$

$$\text{Tsybakov } \tilde{O}(n^{-\frac{1+\kappa}{2+\kappa}}) \tilde{O}(n^{-1/2})$$

E.2 Mammen-Tsybakov Margin Condition

E.2 (Mammen & Tsybakov, 1999). P Margin $\alpha \geq 0$ $c > 0$

$$\mathbb{E}[|2\eta(X) - 1| \cdot \mathbf{1}\{|\eta(X) - 1/2| \leq h\}] \leq ch^\alpha, \quad \forall h > 0.$$

$$\alpha > 0 \text{ SCX}$$

$$\text{SCX NoiseScore}(x_i) = r_i/(\rho(s_i) + \varepsilon) \cdot [1 - C(s_i)] \mathbf{1} - C(s_i) \mathbf{1} s_i$$

$$C(s) \approx P_X(|\eta(X) - 1/2| > \tau_0 \mid X \in s),$$

$$s \quad \eta \approx 1/2$$

E.3 Huber

E.3 (Huber, 1964).

$$F = (1 - \varepsilon)G + \varepsilon H,$$

$$G \rightsquigarrow H \quad \varepsilon \in [0, 1)$$

$$\text{SCX}$$

$$\text{NoiseScore}(x_i) > \tau_{\text{noise}} \Rightarrow x_i \text{ .}$$

$$s \quad \varepsilon(s) = \frac{|\{x_i \in s: \text{NoiseScore}(x_i) > \tau\}|}{|s|}$$

E.4 EM

EM Dawid-Skene - **E** : - **M** :

$$\text{SCX EM 1. } \mathbf{E} : \Pi^{(t)} \text{ } SCX_m^{(t)}(s) \rightsquigarrow \text{ 2. } \mathbf{M} :$$

E.1 (). EM

F. (Active Acquisition)

F.1 MacKay

F.1 (MacKay, 1992).

$$x^* = \arg \max_x H[\theta \mid \mathcal{D}] - \mathbb{E}_{y \sim P(y|x, \mathcal{D})} [H[\theta \mid \mathcal{D} \cup \{(x, y)\}]],$$

H Shannon

$$x^* = \arg \max_x \sigma_n^2(x) = \arg \max_x K(x, x) - \mathbf{k}_n(x)^\top (K_n + \sigma^2 I)^{-1} \mathbf{k}_n(x).$$

F.2 (OED)

- **A-**: $\text{Tr}[(X^\top X)^{-1}]$
- **D-**: $\det(X^\top X)$
- **E-**: $\lambda_{\min}(X^\top X)$

SCX $V(s)$ s D-

$$V(s) \propto \det(\mathbb{E}_{x \in s} [xx^\top]),$$

Fisher

F.3 Bandit Theory (UCB, Thompson Sampling)

UCB (Upper Confidence Bound) (Auer et al., 2002)

$$a_t = \arg \max_{a \in \mathcal{A}} \hat{\mu}_a(t-1) + \sqrt{\frac{2 \log t}{n_a(t-1)}}.$$

SCX $s^* = \arg \max_s V(s)$ **(state-arm) bandit**

$$V(s) = \underbrace{\bar{r}(s) \cdot \rho(s)}$$

F.4 (Acquisition Functions)
(Expected Improvement) (Mockus, 1978)

$$\text{EI}(x) = \mathbb{E}[\max(f(x)-f(x^*), 0)] = (\mu(x)-f(x^*))\Phi(\frac{\mu(x) - f(x^*)}{\sigma(x)})+\sigma(x)\phi(\frac{\mu(x) - f(x^*)}{\sigma(x)}).$$

(Knowledge Gradient) (Frazier et al., 2008)

$$\text{KG}(x) = \mathbb{E}[\max_{x'} \mu_{n+1}(x') \mid x] - \max_{x'} \mu_n(x').$$

$$\text{SCX } V(s) \text{ KG } V(s) \text{ } s \max_m \text{ SCX }_m(s) \text{ KG } \max_{x'} \mu_{n+1}(x') \text{ ---}$$

Proposition 1:

$$\mathbf{1.} \; \mathcal{F} = \{f_1, \ldots, f_M\} \; \Pi : \mathcal{X} \rightarrow \{s_1, \ldots, s_K\} \; P \; \ell$$

$$\forall m, m' : \quad R(f_m) < R(f_{m'}) \quad ()$$

$$s_k$$

$$R_{m'}(s_k) < R_m(s_k) \quad ().$$

$$R(f_m) = \mathbb{E}_{x \sim P}[\ell(f_m(x), f^*(x))] \; R_m(s_k) = \mathbb{E}_{x \sim P(\cdot | s_k)}[\ell(f_m(x), f^*(x))]$$

$$(). \mathcal{X} = [0, 1] \quad f^* : \mathcal{X} \rightarrow \mathbb{R} \quad f_1, f_2$$

$$P_X \sim U[0, 1]$$

$$f^*(x) = \begin{cases} 1, & x \in [0, 0.5) \\ 0, & x \in [0.5, 1] \end{cases}$$

$$f_1(x) = 0.99 - 0.02x$$

$$f_2(x) = \begin{cases} 0.01, & x \in [0, 0.5) \\ 0.99, & x \in [0.5, 1] \end{cases}$$

$$\ell(\hat{y}, y) = (\hat{y} - y)^2$$

$$\begin{aligned} R(f_1) &= \int_0^{0.5} (0.99 - 0.02x - 1)^2 dx + \int_{0.5}^1 (0.99 - 0.02x - 0)^2 dx \\ &\approx \int_0^{0.5} (0.01 + 0.02x)^2 dx + \int_{0.5}^1 (0.99 - 0.02x)^2 dx \\ &= [\dots] \\ &\approx 0.000417 + 0.240417 = 0.240834 \end{aligned}$$

$$\begin{aligned} R(f_2) &= \int_0^{0.5} (0.01 - 1)^2 dx + \int_{0.5}^1 (0.99 - 0)^2 dx \\ &= \int_0^{0.5} (0.99)^2 dx + \int_{0.5}^1 (0.99)^2 dx \\ &= 0.5 \cdot 0.9801 + 0.5 \cdot 0.9801 = 0.9801 \end{aligned}$$

$$R(f_1) = 0.2408 < 0.9801 = R(f_2) \quad f_1$$

$$\bullet \quad s_1 = [0, 0.5), \quad s_2 = [0.5, 1]$$

$$s_1$$

$$\begin{aligned} R_1(s_1) &= \frac{1}{0.5} \int_0^{0.5} (0.99 - 0.02x - 1)^2 dx \approx 2 \times 0.000417 = 0.000834 \\ R_2(s_1) &= \frac{1}{0.5} \int_0^{0.5} (0.01 - 1)^2 dx = 2 \times 0.5 \times 0.9801 = 0.9801 \end{aligned}$$

$$R_1(s_1) < R_2(s_1)$$

$$s_2$$

$$R_1(s_2) = \frac{1}{0.5} \int_{0.5}^1 (0.99 - 0.02x - 0)^2 dx \approx 2 \times 0.240417 = 0.480834$$

$$R_2(s_2) = \frac{1}{0.5} \int_{0.5}^1 (0.99 - 0)^2 dx = 2 \times 0.5 \times 0.9801 = 0.9801$$

$$R_1(s_2) < R_2(s_2)$$

$$\cdot f_1 \ s_2$$

$$f'_1(x) = \begin{cases} 0.99 - 0.02x, & x \in [0, 0.5) \\ 5, & x \in [0.5, 1] \end{cases}$$

$$f'_2(x) = \begin{cases} 0.01, & x \in [0, 0.5) \\ 0.02, & x \in [0.5, 1] \end{cases}$$

$$s_2 \ R_1(s_2) = (1/0.5) \int_{0.5}^1 (5 - 0)^2 dx = 2 \cdot 0.5 \cdot 25 = 25 \ R_2(s_2) = (1/0.5) \int_{0.5}^1 (0.02 - 0)^2 dx = 2 \cdot 0.5 \cdot 0.0004 = 0.0004$$

$$R_1(s_2) = 25 > 0.0004 = R_2(s_2)$$

$$R(f'_1) = 0.000834 \cdot 0.5 + 25 \cdot 0.5 = 12.500417$$

$$R(f'_2) = 0.9801 \cdot 0.5 + 0.0004 \cdot 0.5 = 0.49025$$

$$f'_2 \ s_1 \ \tilde{f}_2 = f'_1, \tilde{f}_1 = f'_2 \ R(\tilde{f}_1) < R(\tilde{f}_2) \ R_{\tilde{f}_2}(s_2) < R_{\tilde{f}_1}(s_2) \ s_2$$

$$P_X(s_1) = 0.9, P_X(s_2) = 0.1$$

		$R(s_1)$	$R(s_2)$	R
f_1	1	100	$0.9 \times 1 + 0.1 \times 100 = 10.9$	
f_2	2	5	$0.9 \times 2 + 0.1 \times 5 = 2.3$	

$$R(f_1) = 10.9 > 2.3 = R(f_2) \ s_1 \ R_1(s_1) = 1 < 2 = R_2(s_1)$$

$$\mathbf{1.1} \text{ (). } R_m \leq R_{m'} \text{ } R_m(s) \leq R_{m'}(s) \text{ } s \text{ } f_m, f_{m'} \text{ } \Delta_{m,m'}(x) = \ell(f_m(x), f^*(x)) - \ell(f_{m'}(x), f^*(x)) \text{ } P_X \text{ } dP_X/dP_X(\cdot|s) \text{ } \psi_{m,m'}$$

$$\Delta_{m,m'}(x) \cdot \frac{dP_X}{dP_X(\cdot|s)}(x) \geq 0, \quad \forall x \in \mathcal{X}, \forall s \in \Pi.$$

Arrow

Arrow Pareto

SCX "" s ""Proposition 1 Arrow

| Arrow | SCX | |————|——| | i | s | | a | f_m | | | R_m | | | | | |

SCX "" Arrow ———

Proposition 2:

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$$

- $H_{\text{learnable}} = \{x : \mathbb{E}[\ell(f(x), y) \mid x] \}$
- $H_{\text{noise}} = \{x : \mathbb{E}[\ell(f(x), y) \mid x] \}$

$$P(H_{\text{noise}}) > 0 \text{ residual-based } \pi_{\text{res}}(x) \propto r_i = \ell(f(x_i), y_i)$$

2. $P_X \ell$ $0 \leq \ell \leq L_{\max}$ B - Max-Residual Sampling: r_i B - State-Level Sampling (SCX): $V(s)$ s

$$\mathbb{E}[\text{res}] \leq \mathbb{E}[\text{state}] - \delta(B),$$

$$\delta(B) > 0 \text{ } B$$

$$\cdot \text{ } s_1, s_2 \text{ } n$$

$$s_1 \quad y_i = f^*(x_i) + \epsilon_i \quad \epsilon_i \sim N(0, \sigma_{\text{low}}^2) \quad - \quad f \quad s_1 \quad \mathbb{E}[f(x) \mid x \in s_1] = f^*(x) + \mu$$

$$\mu \gg 0$$

$$s_2 \quad y_i = + \eta_i \quad \eta_i \sim N(0, \sigma_{\text{high}}^2) \quad \sigma_{\text{high}}^2 \gg \sigma_{\text{low}}^2 \quad - \quad f \quad s_2 \quad \mathbb{E}[f(x) \mid x \in s_2] =$$

•

$$s_1$$

$$r_i^{(1)} = (f(x_i) - y_i)^2 = (\mu - \epsilon_i)^2 = \mu^2 - 2\mu\epsilon_i + \epsilon_i^2.$$

$$\mathbb{E}[r^{(1)}] = \mu^2 + \sigma_{\text{low}}^2.$$

$$s_2$$

$$r_i^{(2)} = (f(x_i) - y_i)^2 = \eta_i^2.$$

$$\mathbb{E}[r^{(2)}] = \sigma_{\text{high}}^2.$$

$$\bullet \quad \mu^2 + \sigma_{\text{low}}^2 < \sigma_{\text{high}}^2 \quad s_2 \quad s_1$$

$$\textbf{Max-Residual Sampling} \quad \bullet \quad B \quad \sigma_{\text{high}}^2 \gg \mu^2 + \sigma_{\text{low}}^2 \quad s_2$$

$$\bullet \quad s_2 \text{ — } s_1 \quad \mu^2 \quad O(\sigma_{\text{low}}^2/n_{s_1})$$

$$\bullet \quad \text{Max-Residual} \quad B \quad s_2$$

$$\mathbb{E}[\text{res}] = O\left(\frac{\sigma_{\text{low}}^2}{n_{s_1}}\right) \cdot \left(\frac{n_{s_1}}{n_{s_1} + n_{s_2}}\right)^B \rightarrow 0 \quad \text{as } B \rightarrow \infty$$

$$\frac{n_{s_1}}{n_{s_1} + n_{s_2}} < 1$$

$$\bullet \quad B_{\text{res}} \quad \text{Max-Residual} \quad s_1 \quad B_{\text{state}} \quad \text{State-Level Sampling} \quad s_1 \quad \text{Max-Residual} \quad s_2 \quad B_{\text{res}} \ll B_{\text{state}}$$

$$\text{- Max-Residual: } \mathbb{E}[\mathcal{L}_{\text{res}}] \approx \frac{\sigma_{\text{low}}^2}{n_{s_1} + B_{\text{res}}} + \sigma_{\text{high}}^2 \quad \text{- State-Level: } \mathbb{E}[\mathcal{L}_{\text{state}}] \approx \frac{\sigma_{\text{low}}^2}{n_{s_1} + B_{\text{state}}} + \sigma_{\text{high}}^2$$

$$\mathbb{E}[\mathcal{L}_{\text{res}}] - \mathbb{E}[\mathcal{L}_{\text{state}}] \approx \sigma_{\text{low}}^2 \left(\frac{1}{n_{s_1} + B_{\text{res}}} - \frac{1}{n_{s_1} + B_{\text{state}}} \right) > 0$$

$$\mathbb{E}[\text{state}] - \mathbb{E}[\text{res}] > 0$$

SCX

2.1 (). SCX $V(s) = \bar{r}(s) \cdot \rho(s) \cdot L(s) \cdot [1 - D(s)]$ $L(s)$ s SCX

$$\mathbb{E}[\text{SCX}] \leq B \cdot \epsilon_{\text{est}},$$

$$\epsilon_{\text{est}} = P(L(s) \text{ Max-Residual})$$

$$\mathbb{E}[\text{res}] \geq B \cdot P(x \in H_{\text{noise}} \mid).$$

$$P(x \in H_{\text{noise}} \mid) \gg \epsilon_{\text{est}} \text{ SCX Max-Residual}$$

Proposition 3:

SCX f_m

$$w_m(x) \propto \exp\left(-\alpha \sum_x \gamma_s(x) \hat{R}_m(s)\right),$$

$$\gamma_s(x) = \mathbf{1}\{x \in s\}$$

$$w_m^{\text{global}} \propto \exp(-\alpha \hat{R}_m).$$

3. $w_m(x) w_m^{\text{global}}$

$$\mathbb{E}_{x \sim P} \left[\sum_m w_m(x) \ell(f_m(x), y) \right] \leq \mathbb{E}_{x \sim P} \left[\sum_m w_m^{\text{global}} \ell(f_m(x), y) \right],$$

ΠY

.

$$w_m(s) = \frac{\exp(-\alpha \hat{R}_m(s))}{\sum_{m'} \exp(-\alpha \hat{R}_{m'}(s))}.$$

$$w_m^{\text{global}} = \frac{\exp(-\alpha \hat{R}_m)}{\sum_{m'} \exp(-\alpha \hat{R}_{m'})}.$$

Step 1: Gibbs . $x \in s$ s $w_m(s)$

$$J_s(w) = \sum_m w_m \hat{R}_m(s) + \frac{1}{\alpha} \sum_m w_m \log w_m.$$

KKT $w_m(s)$ w' w_m^{global}

$$\sum_m w_m(s) \hat{R}_m(s) + \frac{1}{\alpha} \sum_m w_m(s) \log w_m(s) \leq \sum_m w_m^{\text{global}} \hat{R}_m(s) + \frac{1}{\alpha} \sum_m w_m^{\text{global}} \log w_m^{\text{global}}.$$

Step 2: . $x \in s$

$$\mathbb{E} \left[\sum_m w_m(s) \ell(f_m(X), Y) \mid X \in s \right] + \frac{1}{\alpha} H(w(s)) \leq \mathbb{E} \left[\sum_m w_m^{\text{global}} \ell(f_m(X), Y) \mid X \in s \right] + \frac{1}{\alpha} H(w^{\text{global}}),$$

$H(w) = -\sum_m w_m \log w_m$ Shannon

Step 3: . Jensen

$$H(w^{\text{global}}) \leq \mathbb{E}_s[H(w(s))],$$

$$-H \quad H(\mathbb{E}[w(s)]) \geq \mathbb{E}[H(w(s))]$$

$$\mathbb{E} \left[\sum_m w_m(s) \ell(f_m(X), Y) \right] \leq \mathbb{E} \left[\sum_m w_m^{\text{global}} \ell(f_m(X), Y) \right].$$

. $\hat{R}_m(s) = \hat{R}_m$ s

3.1 (Bound). $\ell \in [0, L_{\max}]$

$$\mathbb{E} \left[\sum_m w_m(s) \ell(f_m(X), Y) \right] \leq \mathbb{E} \left[\sum_m w_m^{\text{global}} \ell(f_m(X), Y) \right] - \frac{1}{\alpha} \cdot I(s; w),$$

$$I(s; w) = \mathbb{E}_s[\text{KL}(w(s) \| w^{\text{global}})]$$

:

$$\begin{aligned} & \mathbb{E} \left[\sum_m w_m^{\text{global}} \ell(f_m(X), Y) - \sum_m w_m(s) \ell(f_m(X), Y) \right] \\ &= \mathbb{E} \left[\sum_m (w_m^{\text{global}} - w_m(s)) \ell(f_m(X), Y) \right] \\ &= \mathbb{E} \left[\sum_m (w_m^{\text{global}} - w_m(s)) (\ell(f_m(X), Y) - \hat{R}_m(s)) \right] + \mathbb{E} \left[\sum_m (w_m^{\text{global}} - w_m(s)) \hat{R}_m(s) \right] \\ &\geq \mathbb{E} \left[\sum_m (w_m^{\text{global}} - w_m(s)) \hat{R}_m(s) \right] \quad (\ell - \hat{R}) \end{aligned}$$

Gibbs

$$\sum_m (w_m^{\text{global}} - w_m(s)) \hat{R}_m(s) = \frac{1}{\alpha} \sum_m w_m(s) \log \frac{w_m(s)}{w_m^{\text{global}}} = \frac{1}{\alpha} \text{KL}(w(s) \| w^{\text{global}}).$$

Bound $-I(s; w) > 0$

3.1 $R_m(s)$

. $n \hat{R}_m(s)$

3.1.1 (). $\ell \in [0, L_{\max}]$ $s \ n_s \ \delta > 0 \ 1 - \delta$

$$|\hat{R}_m(s) - R_m(s)| \leq L_{\max} \sqrt{\frac{\log(2/\delta)}{2n_s}}.$$

. Hoeffding

. $|\hat{R}_m(s) - R_m(s)| \leq \epsilon \ 1 - \delta$

$$n_s \geq \frac{L_{\max}^2 \log(2/\delta)}{2\epsilon^2}.$$

- $s \ n_s \ R_m(s)$ - SCX $\rightarrow \rightarrow$

. M Bonferroni

$$n_s \geq \frac{L_{\max}^2 \log(2M/\delta)}{2\epsilon^2}.$$

3.1.2 (). $\hat{R}_m(s)$

$$\mathbb{E}[|\hat{R}_m(s) - R_m(s)|] \geq \frac{\sigma(s)}{2\sqrt{n_s}},$$

$\sigma^2(s) = \text{Var}(\ell(f_m(X), f^*(X)) \mid X \in s)$

. Le Cam Fano

3.2

.

3.2.1 (). $\mathcal{X} \ \Pi^* = \{s_1^*, \dots, s_K^*\} \ s_k^* \ P_{Y|X} \ P(Y \mid X) = P(Y \mid s_k^*) \ x \in s_k^*$

3.2.1 (). 1. $\mathcal{X}$ Riemann $\mathcal{M}$ 2. $P_{Y|X} \ \mathcal{M} \ \mathcal{J}$ 3. $K_h(x, x') = \exp(-\|x - x'\|^2/h)$
 $h \rightarrow 0 \ h \cdot n^{1/(d+4)} \rightarrow \infty$

$$\hat{\Pi}_n$$

$$d_{\text{Hausdorff}}(\partial \hat{\Pi}_n, \mathcal{J}) \xrightarrow{P} 0, \quad n \rightarrow \infty,$$

$$d_{\text{Hausdorff}} \text{ Hausdorff } \partial \hat{\Pi}_n$$

$$\cdot \text{ von Luxburg et al. (2008) } Y$$

$$\cdot \text{ SCX } K \text{ SCX } \longrightarrow n \rightarrow \infty$$

$$K \cdot K$$

$$K^* = \arg \min_K \sum_{k=1}^K \sum_{x_i \in \hat{s}_k^{(K)}} \ell(\bar{y}_{\hat{s}_k^{(K)}}, y_i) + \lambda K,$$

$$\bar{y}_{\hat{s}_k^{(K)}}$$

$$\mathbf{3.2.2} \text{ (). } K^* \ n \ \hat{K}_n = K^* \ \ell \ \lambda \ \lambda \gg \sigma^2 \log(n)/n$$

3.3 State-wise vs. Point-wise Regret Bound

$$\cdot \text{ SCX } m^*(x) = \arg \min_m R_m(s(x)) \ m^{**}(x) = \arg \min_m \mathbb{E}[\ell(f_m(x), y) \mid x]$$

$$\cdot \text{ regret}$$

$$\text{Regret}_{\text{state}}(n) = \mathbb{E} \left[\sum_{m=1}^M \mathbf{1}\{m^*(X) = m\} \ell(f_m(X), Y) \right] - \mathbb{E} \left[\min_m \ell(f_m(X), Y) \right].$$

$$\mathbf{3.3.1 \text{ (Regret Bound). }} \ [0, L_{\max}] \ \epsilon\text{-}$$

$$\mathbb{E}_{x \in s}[|\eta(x) - \eta(s)|] \leq \epsilon, \quad \forall s \in \Pi,$$

$$\eta(x) = \mathbb{E}[Y \mid X = x] \eta(s) = \mathbb{E}[Y \mid X \in s]$$

$$\text{Regret}_{\text{state}} \leq 2L_{\max} \cdot \epsilon.$$

•

$$\begin{aligned}
\text{Regret}_{\text{state}} &= \mathbb{E} \left[\min_m \mathbb{E}[\ell(f_m(X), Y) \mid X \in s] - \min_m \ell(f_m(X), Y) \right] \\
&\leq \mathbb{E} \left[\max_m |\mathbb{E}[\ell(f_m(X), Y) \mid X \in s] - \ell(f_m(X), Y)| \right] \\
&\leq \mathbb{E} [|\mathbb{E}[Y \mid X \in s] - Y|] \cdot L'_{\max} \quad (\text{Lipschitz}) \\
&\leq 2L_{\max} \cdot \epsilon.
\end{aligned}$$

• $\epsilon \rightarrow 0$ regret

• $\epsilon_n \ n \ \epsilon_n = O(n^{-1/(d+2)}) \ d$

$$\text{Regret}_{\text{state}}(n) = O(n^{-1/(d+2)}).$$

3.4

• H_{noise}

3.4.1 (). $s \ \Delta$ -

$$|\bar{r}(s) - \bar{r}(s')| > \Delta \quad |L(s) - L(s')| > \Delta, \quad \forall s' \neq s,$$

$L(s)$

3.4.1 ().

1. ℓ
2. $s \ n_s \geq c \cdot \log(M/\delta)/\Delta^2$
3. $K = o(n/\log n)$

$\mathcal{A}$

$$P(\mathcal{A}) \geq 1 - \delta.$$

•

$H_0 : s \in H_{\text{learnable}}$ vs. $H_1 : s \in H_{\text{noise}}$ Chernoff-Stein $\exp(-n_s \cdot \text{KL}(P_1 \| P_0))$

4.1 SCX PAC-Bayes

. s ”” P_s ”” Q_s

$$Q_s(m) = \frac{\exp(-\alpha R_m(s))}{\sum_{m'} \exp(-\alpha R_{m'}(s))}.$$

4.1 (PAC-Bayes) . s Q_s $1 - \delta$

$$\mathbb{E}_{m \sim Q_s}[R_m(s)] \leq \mathbb{E}_{m \sim Q_s}[\hat{R}_m(s)] + \sqrt{\frac{\text{KL}(Q_s \| P_s) + \log(n_s/\delta)}{2n_s - 1}}.$$

SCX . $\alpha \rightarrow \infty$ Q_s $\delta_{m^*(s)}$ PAC-Bayes

$$R_{m^*(s)}(s) \leq \hat{R}_{m^*(s)}(s) + \sqrt{\frac{\log M + \log(n_s/\delta)}{2n_s - 1}}.$$

SCX ””

PAC-Bayes vs. PAC-Bayes. Q

$$R(Q) \leq \hat{R}(Q) + \sqrt{\frac{\text{KL}(Q \| P) + \log(n/\delta)}{2n - 1}},$$

$\text{KL}(Q \| P)$ $\text{KL}(Q \| P) \geq \mathbb{E}_s[\text{KL}(Q_s \| P_s)]$ PAC-Bayes PAC-Bayes

4.2 Information Bottleneck Principle

(Tishby, 1999). $\tilde{X}$ $I(\tilde{X}; Y)$ $I(\tilde{X}; X)$

$$\min_{P(\tilde{x}|x)} I(\tilde{X}; X) - \beta I(\tilde{X}; Y).$$

SCX $\tilde{X} = s(X)$ $\tilde{X}$

4.2 (). $|\tilde{\mathcal{X}}| = K$ SCX Π^*

$$\Pi^* \in \arg \min_{\Pi: |\Pi|=K} I(X; \tilde{X}) - \beta I(\tilde{X}; Y),$$

$\tilde{X} = s(X) \mid X \in s$
 $\cdot I(X; \tilde{X}) = H(\tilde{X}) - \tilde{X} \mid X \mid H(\tilde{X}) \leq \log K \mid I(\tilde{X}; Y) \mid \mathbb{E}_s[H(Y \mid X \in s)]$ SCX ” Y
 „

SCX . ””SCX

$$\min_{\Pi} I(X; \tilde{X} \mid \mathcal{F}) - \beta I(\tilde{X}; Y \mid \mathcal{F}),$$

$\mathcal{F} = \{f_1, \dots, f_M\}$ ———
 . SCX

4.3

. SCX (Wald, 1950)
 1. Θ 2. $\mathcal{A}$ f_m 3. $L(\theta, a)$
 . $d : \mathcal{X} \rightarrow \mathcal{A}$ SCX -

$$d(x) = \arg \min_m \mathbb{E}[\ell(f_m(x), f^*(x)) \mid x \in s(x)].$$

4.3 (). Π

$Y \perp X \mid s(X)$ - $d(x)$ admissible——
 . Lehmann-Scheffe $T(X)$ $T(X)$
 $s(X)$ X -
 SCX . $\pi(\theta)$

$$d_{\text{Bayes}}(x) = \arg \min_a \mathbb{E}_{\theta \sim \pi(\cdot \mid x)}[L(\theta, a)].$$

SCX $w_m(x)$

$$w_m(x) \approx P(\theta_m \mid x).$$

Neyman-Pearson . SCX Neyman-Pearson s $H_0 : x$ vs. $H_1 : x$

$$\Lambda(x) = \frac{P_{\text{learnable}}(x)}{P_{\text{noise}}(x)} \leq \tau.$$

SCX NoiseScore(x_i) $\Lambda(x_i)$

s			Def 1
$R_m(s)$	f_m	s	Def 1
$SCX_m(s)$	f_m	s	Def 2
$V(s)$	s		Def 3
$\bar{r}(s)$	s		Def 3
$\rho(s)$	s		Def 3
$L(s)$	s		Def 3
$D(s)$	s		Def 3
$C(s)$	s	/	
$N(s)$	s		Def 3
$w_m(x)$	f_m	x	
$\gamma_s(x)$			
$\eta(x)$	$\mathbb{E}[Y \mid X = x]$		A.2
ℓ			Def 1
τ			Def 2
α	softmax		
λ			
$\phi_i(v)$	i	<u>Shapley</u>	C.1

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